

Overview:

- My vision for the first prototype of our proposed Chameleon Inventory Management software is that the user will open our web app and once on the home page they can either choose to sign in to access the staff area or browse our catalogue, once signed in the user can browse the catalogue, add/remove items from stock, scan QR codes to update status and stock, or view transaction logs.

Features to be developed:

- F-1: Role based access control
Role based access control is essential in securing the web app, and ensuring that non-staff users cannot edit or read data, and to also ensure that teachers cannot edit or use the functions to change stock or sign out stock, only browse the catalogue.
- F-2: Stock signing in process
This is essential for allowing the correct staff to check stock back into storage or one of the other locations in HQ.
- F-3: Stock signing out process
This feature is essential to checking stock out for internal shows, or external hires.
- F-4: Add/Remove stock
These features are essential for adding new stock to the system or deleting lost or damaged items from the system.
- F-5: Browse stock catalogue
This feature allows HQ staff and teachers to browse and find costumes.
- F-6: Filtering and searching
This allows costumes to be filtered by identifying features like colour/size, and or search by name.
- F-7: Transaction logs
This allows admin staff to oversee the transactions and work being performed by the HQ staff. And ensure that items are not lost.
- F-8: Tool/Tech/Prop/Costume stock lists
This allows everyone to see what stock we have and allows better and more accurate/efficient working.
- F-9: QR code identification system.

This allows staff to just scan a costume to bring up its page instead of searching.

Technologies:

- Programming languages:

Java: Java should be used due to the OOP module all students undertook in year 1 of this course. All students have had experience with the language and have spent time learning and coding in the language which aligns our team's skill and allows us to be realistic in our goals within our three week timeframe. Java is also a very strong language for use in the backend of our system.

SQL: All students gained hands-on experience with using SQL and creating queries earlier this year so using this language will ensure that we can use databases efficiently.

- Database:

PostgreSQL Relational Database: A PostgreSQL relational database should be used as it allows us to separate data into tables so we can store customer details and reports separately. Another reason is that to align with the group's ability as earlier in year 2 we all had exposure to using and building PostgreSQL databases in the database and web programming module.

- Frameworks:

Spring Boot (Backend): Using Spring Boot is beneficial as it is pure java a language we are all experienced in so should allow us to work without hindrance, It has easy database integration, built in REST support, and is an industry standard. It also provides security using the spring data JPA / Hibernate.

React (Frontend): By using React we are developing real world skills in an industry standard technology. It integrates well with Spring Boot, allows us to build fast interactive scalable web applications, and user interfaces for mobile devices.

- Overall technology stack:

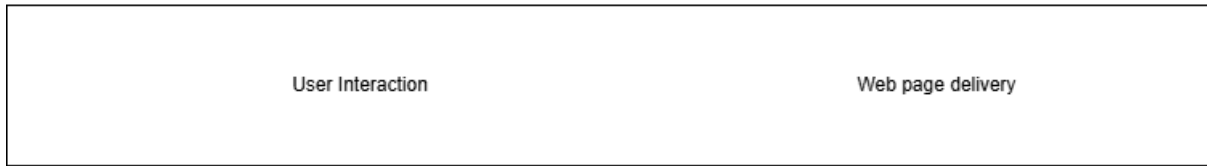
Frontend: React, HTML/CSS, JavaScript, Leaflet.

Backend: Spring Boot, Java.

Database: PostgreSQL.

General Architectural design:

User interface:



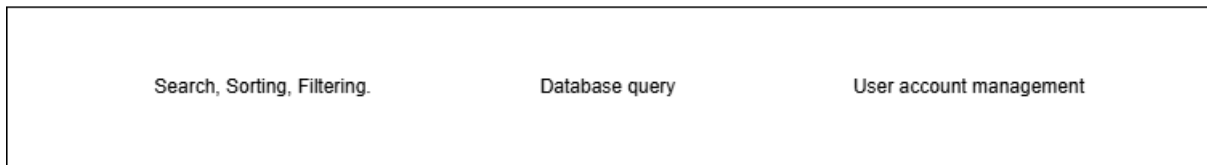
User interface management:



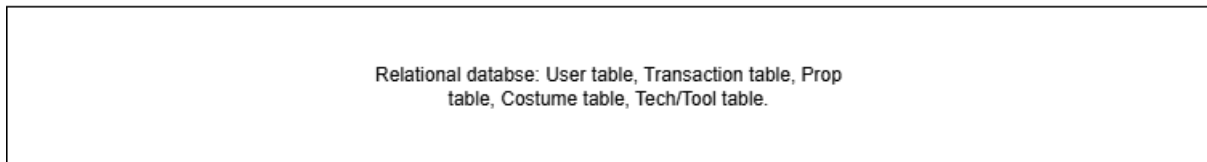
Data Handling:



Basic shared services:

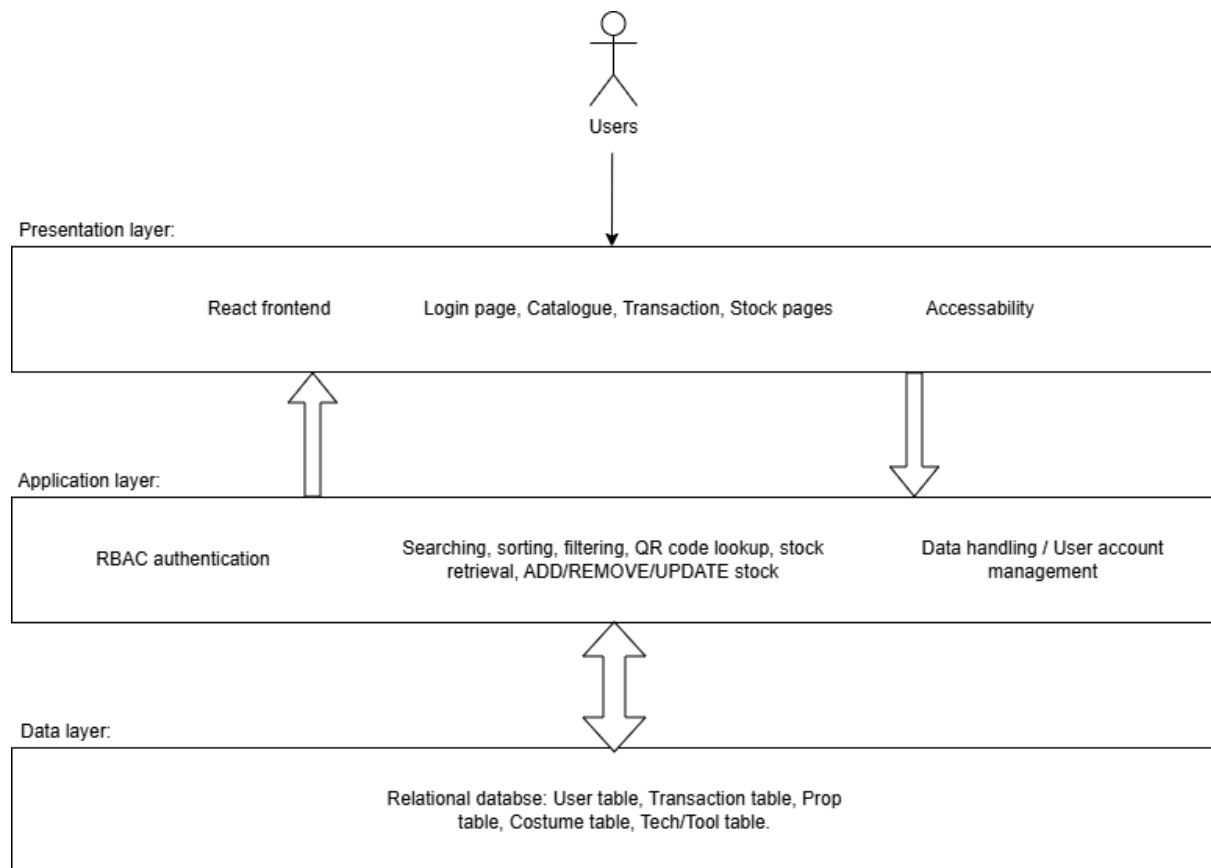


Database:



<u>Layer</u>	<u>Explanation:</u>
User Interface	This layer takes the input from the user and also delivers the UI / web pages which is why User interaction, and Web page delivery are in this layer.
UI management	This layer is where the UI is generated and where the user and their input is authenticated/authorised.
Data Handling	Adding/Removing/Updating stock, stock retrieval, QR code lookup, happen in the same layer so that the data can be retrieved for the updating and lookup.
Basic Shared Services	These services can be used at all stages of the system so they come under the basic shared services layer..

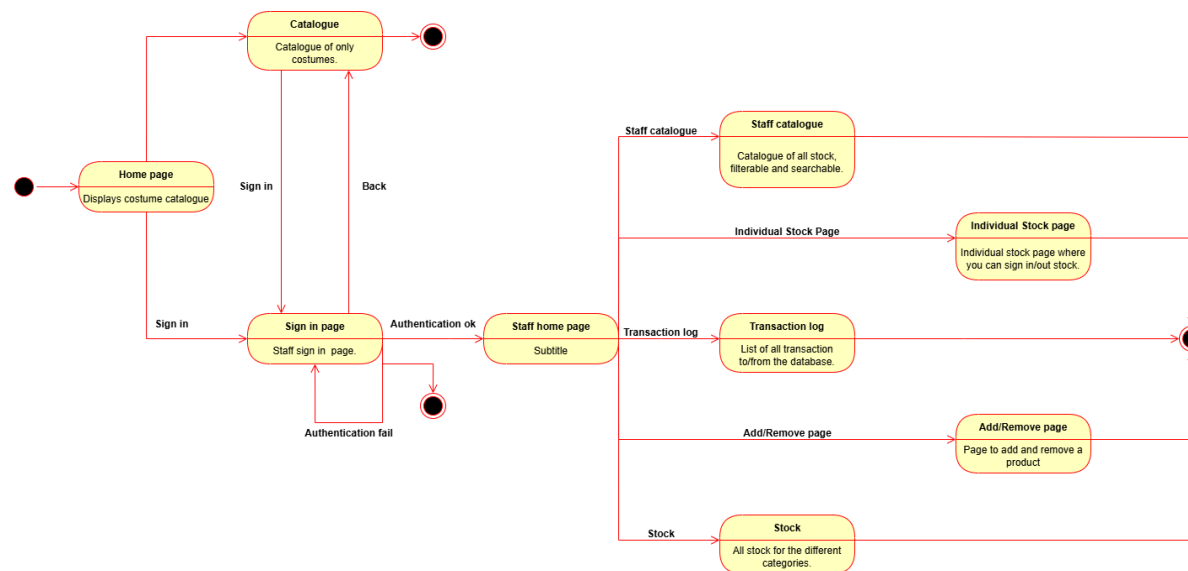
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Object:	Description:
React frontend.	The front end of the system.
Login page.	Page for staff to sign in.
Catalogue page.	Catalogue page for viewing costumes.
Transaction page.	Page for viewing list of transactions.
Stock page.	Page for viewing stock lists.
Individual page.	Page for information on individual stock items.
Add/Remove page.	Page for adding or removing to/from the stock.
Accessibility features.	The place where users can activate/deactivate accessibility features.
RBAC authentication.	The built-in service where users are authenticated based on their roles.
Searching, sorting, filtering.	The built in search, filtering, and sorting.
QR code lookup.	The place to scan the QR code to bring up the specific stock item.
Stock retrieval.	The service that retrieves stock items.

Data handling.	The service to handle data transfer and sending.
User account management.	The part of the system where accounts are managed.
Database.	The data store for the service.

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State:	Description:
Home page.	The landing page for all users, where the costume catalogue is displayed.
Catalogue.	The display of all costumes, filterable, and searchable.
Sign in page.	The page for Starlight staff to sign in to use the application.
Staff home page.	The home page for authenticated users, Displays the staff catalogue.
Staff catalogue.	The display of all items, including tech, tools, props, and costumes.
Transaction log.	The display of all database transactions.
Stock page.	The display of all stock, sorted into sections, list format.
Add/Remove page.	The page for adding or removing specific stock from the system.
Individual stock page.	The page for each individual costume where you can sign them in or out.

Stimulus:	Description:
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Sign in.	The user presses the sign in button.
Back.	The user goes back to the home page from the sign in page.
Authentication OK.	The user is authenticated as a staff member and let in to the site.
Authentication FAIL.	The user is not authenticated and asked to authenticate themselves or quit the application.
Staff catalogue.	The user presses the button to view the staff catalogue.
Transaction log.	The user chooses to view the transaction log page.
Stock.	The user presses the button to view the stock takes.
Individual stock page.	The user scans a QR code/or searches the item to open the individual stock page.
Add/Remove page.	The user chooses to open the page to add or remove to/from the stock.

High level feature designs:

Base classes:

Base Classes

Costume
+ costumeID: String + name: String + size: Character + quantity: Integer + colour: String + status: ENUM + locationCode: String + lastUpdated: DateTime + notes: String
GET/SET

Prop
+ propID: String + name: String + variant: String + quantity: Integer + locationCode: String
GET/SET

Tech
+ techID: String + name: String + quantity: Integer + locationCode: String + condition: ENUM + assignedTo: String + ownedBy: String
GET/SET

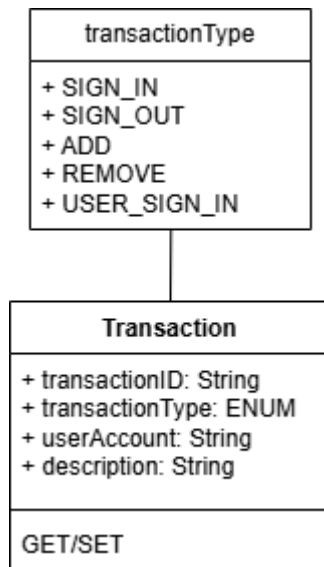
Tool
+ toolID: String + name: String + quantity: Integer + locationCode: String + condition: ENUM + assignedTo: String + ownedBy: String
GET/SET

ENUMs

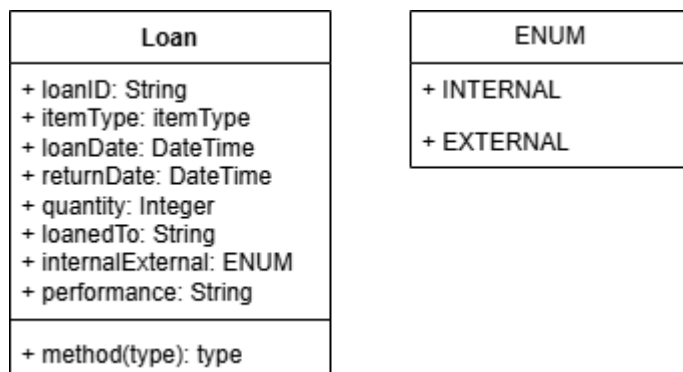
Status
+ IN_REPAIR + STORAGE + LAUNDRY + INTERNAL_HIRE + EXTERNAL_HIRE + REMOVAL + LOST

Condition
+ NEW + USED + UNKNOWN + UNOPENED + WORKING + NEEDS_REPAIR

F-7:



Loan class:



Signing In process:

1. Either search for the specific item in the catalogue or scan the QR code to bring up the item's individual page.
2. Press the "Sign item in." button.
3. Then fill out the sign in form consisting of: quantity, status, returnedFrom. And press enter.

Signing out process:

1. Either search for the specific item in the catalogue or scan the QR code to bring up the item's individual page.
2. Press the "Sign item out." button.
3. Then fill out the sign out form consisting of: quantity, loanedTo, showLoan, expectedReturnDate, Internal/external loan. And press enter.

